

Khushi Upadhyay

Software Engineer

[Gmail](#) | (469) 509-0528 | [Canada \(Citizen\)](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Summary

Software Engineer with 3 years of experience designing and delivering scalable, secure, and high-performance applications. Skilled in full-stack development, cloud platforms, microservices, and CI/CD automation. Strong background in API development, distributed systems, and data-driven solutions, with a proven ability to collaborate across teams and adapt quickly to evolving technical and business requirements.

Technical Skills

- **Programming, Backend & Frontend Development:** Python, Java, JavaScript, TypeScript, C#, SQL, PL/SQL, Shell Scripting, FastAPI, Django, Express.js, Spring Boot, RESTful APIs, GraphQL, WebSockets, microservices, serverless architecture, React.js, Redux, Next.js, Angular, TypeScript UI development, responsive web design, component-based architecture, Android development using Java and Android Studio
- **Databases & Data Management:** PostgreSQL, MongoDB, AWS RDS, SQL optimization, dynamic SQL, schema design, data validation, financial reporting, transaction reconciliation
- **Cloud Platforms:** AWS (EKS, ECS, Lambda, SQS, SageMaker), Azure (Azure OpenAI, App Service, Functions, Service Bus), GCP
- **DevOps, CI/CD & Infrastructure:** Docker, Kubernetes, GitHub Actions, Jenkins, ArgoCD, Terraform, build, deployment, and release pipelines
- **Messaging, Streaming & Background Processing:** Kafka, Celery, Redis, asynchronous job queues, schedulers, event-driven processing
- **AI, Machine Learning & Automation:** Hugging Face Transformers, BERT, Azure OpenAI, Scikit-learn (KNN, SVM, Random Forest), text classification, anomaly detection, AI-powered workflow automation
- **Testing, QA & Monitoring:** PyTest, Selenium, Jest, React Testing Library, SonarQube, Prometheus, Grafana, AWS CloudWatch, Azure Monitor
- **Authentication, Security & Compliance:** OAuth 2.0, JWT, RBAC, PCI-DSS compliant financial systems, data encryption

Professional Experience

Software Engineer - MetLife Inc.

08/2025 – Present | Remote, USA

- Worked on AI-Powered Claims Automation Platform, developing a Python FastAPI backend with PostgreSQL on AWS RDS, implementing scalable claim processing workflows, and collaborating with product, actuarial, and finance teams during requirement-gathering workshops.
- Built secure REST and GraphQL APIs for /claims, /policies, /accounts, and /insights endpoints, implementing authentication, authorization, and data encryption on AWS Lambda, ensuring security compliance, reducing manual claim errors by 30%, while integrating seamlessly with React frontend.
- Developed React TypeScript dashboards with Redux, WebSockets, and GraphQL subscriptions, collaborating with data science team for integrating Hugging Face Transformers (BERT) on AWS SageMaker for automated claim text classification, improving claim categorization accuracy by 35%.
- Designed microservices architecture with Celery and Redis for asynchronous claim ingestion and processing on AWS ECS, integrating internal insurance APIs, reducing latency by 25%, improving throughput by 28%, and maintaining auditability and operational efficiency.
- Managed CI/CD pipelines and cloud deployment, using GitHub Actions, Docker, and AWS EKS, collaborating closely with DevOps engineers, implementing monitoring with CloudWatch and Prometheus, achieving 99.8% uptime and reducing incident resolution time by 20%.

Software Engineer - Quarterhill Inc.

07/2024 – 08/2025 | TX, USA

- Developed backend for riteOSS Payment Plan system using Java, Python, and PL/SQL, enabling flexible toll payment schedules, while building scalable REST APIs in a microservices architecture deployed via AWS Lambda and automated with GitHub Actions.
- Designed Python-based automation tools using ML models for transaction pattern analysis and anomaly detection, partnered with QA and support teams to optimize high-volume schedulers processing 1K+ transactions per cycle, improving operational efficiency and reliability.
- Wrote complex SQL queries and reporting solutions for transaction validation, enhancing data accuracy, compliance, and financial reporting, while developing secure dynamic SQL workflows and checkprint functionality for refunds via check, credit card, bank transfer, and cash.
- Automated repetitive workflows using Python and shell scripting, reducing manual effort by 40%, and implemented CI/CD pipelines through GitHub Actions and shell scripts, cutting release time by 70% and ensuring reliable, consistent deployments across environments.
- Delivered secure, compliant financial data processing solutions by integrating backend services, automated reporting, anomaly detection, scheduler optimization, and multi-channel refund functionalities, ensuring operational efficiency, regulatory adherence, and high reliability for payment and transaction systems.

Frontend Developer - YuRide

09/2022 – 12/2022 | Toronto, Canada

- Developed and designed responsive frontend components for YuRide's ride-booking app using ReactJS and Java (Android Studio), enabling real-time ride management and seamless user experience for over 100 users across web and Android platforms.
- Improved UI performance and application responsiveness by integrating REST APIs, optimizing frontend logic, and reducing load times by approximately 15%, ensuring smoother interactions and faster access to ride-booking features across multiple platforms.

Web Developer - Excel Lassonde, York University

08/2021 – 09/2022 | Toronto, Canada

- Collaborated with a team to design and develop responsive web pages using HTML, CSS, and JavaScript, enhancing user experience and ensuring cross-browser compatibility while following modern front-end development best practices.
- Implemented Single Sign-On (SSO) authentication and automated email reminders, which streamlined user access, improved session attendance by approximately 25%, and optimized communication workflows between users and the platform.
- Enhanced site usability and performance by optimizing layouts, implementing modern front-end techniques, and managing tasks efficiently in an Agile environment using Git version control for timely delivery and effective team collaboration.

Education

Bachelor of Computer Engineering (Honors) | Certificate in Technology Entrepreneurship
York University, Toronto, Canada

09/2019 – 04/2024

Certificates

- AWS CloudOps Engineer Associate (SOA-C03) (In Progress)
- AWS Generative AI Applications Professional Certificate (Aug 2025)
- Microsoft Azure Fundamentals (AZ-900) (Dec 2024)

Projects

AI-Powered Portfolio Chatbot | Next.js, React, Azure OpenAI, Terraform

- Developed a responsive full-stack portfolio featuring an AI chatbot using Azure OpenAI, built with Next.js and React, and deployed via Azure App Service with Terraform for scalable, reproducible infrastructure.

Predictive Analytics for Human Activity Recognition | Python, Scikit-learn, Pandas

- Designed machine learning models (KNN, SVM, Random Forest) to classify human activities from Wi-Fi CSI data, preprocessing 3D signals and achieving ~87% accuracy while optimizing performance for real-time smart home applications.